Arthur: Signal Research Plan

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Where we are

On 15 years of USD/JPY bars from M1 to daily, B, S, Theta and Mu show no direction edge: no rule beat the always-up rate, so the 55% goal (H5) is not met at any horizon. The only structure found is a tiny volatility-size signal in Theta and S, and moves 2–4× larger around scheduled releases. The 2025 sawtooth in B and S came from the math, not the market.

The robustness review is finished, apart from the OANDA live account (item 14). Signal work comes before trading work. Until a signal shows an edge net of costs, trading stays buy-then-close on the demo account, with `ALLOW_LIVE = False`.

What we found

Window size. Each run fetches `8 × zone` horizon bars, and the first 19 are warm-up because `ECO_ROLL = 20`. At zone 3 that is 24 bars, so only about 5 are real. The 2025 files had 120 to 390 bars, roughly zone 15 to 48. Zone changes how much history you see, not the shape of B and S.

Old vs current B and S. Running the Oct 2025 `transform_signal.py` on the saved bars reproduces the 2025 signal files exactly. Both versions start from the same inputs: X (volume-signed price pressure, smoothed and z-scored) and Y (RSI recentred to between -1 and 1), made orthogonal by Gram-Schmidt.

	2025 math	Current math (commit 019dadc)
S built from	X and Y	-X and -Y
B − S	`(σB + σS − 1)(X − Y)`, never negative	signed, roughly linear in X and Y
Margin	roughly the squared X−Y gap, z-scored over 20 bars, flipped (`MREG = -1`)	a signed momentum-like mix
Shape	sawtooth: ramps up, sharp drops	no sawtooth

On your Sept 2025 file, the old Margin correlates 1.00 with the negative z-score of $(X - Y)^2$.

Random-walk test. The 2025 math was run on your real Sept 25 bars (305 of them) and on 30 random walks with random volume.

Margin statistic	Real data, 2025 math	Random walks, 2025 math	Real data, current math
Skew	-1.28	-1.44	0.04
Share of sharp drops	13.2%	14.3%	5.7%

The sawtooth appears just as strongly on noise. Squaring a gap and z-scoring it over 20 bars turns any data into ramps and drops, so the shape tells us nothing about the market.

Verdicts on the hypotheses

H5 failed: no horizon from M1 to daily reached 55%, and none beat the always-up rate for the same bars. The other hypotheses stand as follows.

#	Verdict	Evidence
H1	Partly	The old squared gap was not isolated. The current Theta and S add 0.03–0.19 points of R^2 on M1–M30 and 0.07–0.15 points on H1/H4, on top of trailing volatility and hour of day. Too small to use
H2	Not tested on its own	The current B−S sign hit 48.7–51.0% on H1, H4 and daily, below always-up
H3	Rejected for direction	B−S>0 and Mu contrarian match or trail always-up at every horizon. Theta×Mu regions, logistic regression and clusters also sit near 50% out of sample
H4	Not isolated	With no direction signal to explain, the comparison against a plain momentum baseline was not run
H5	Failed	Best out-of-sample result: Mu contrarian 51.2% on daily bars against 52.9% always-up

How the tests were run

Every result above follows the same protocol, so the numbers can be compared and repeated.

- **Rules frozen first.** Each rule and its pass criteria were written down before the run (EVENT_RULES.md for the calendar tests, LONG_HORIZON_RULES.md for H1/H4/daily). A lead needed a hit rate above 54%, more than 80% of years above 50%, and at least 1 point over always-up.
- **Benchmark is always-up, not 50%.** USD/JPY rose in 50.4% of bars in 2025, 52.4% in 2026, 51.4% (H1), 52.4% (H4) and 52.9% (daily) over 2012 to July 2026. A rule that hits 51% below those numbers has no edge.
- **Costs.** Median spread 1.6 pips (1.9–2.2 around releases). Swap (overnight interest) is not included.
- **Bars.** Built from mid prices like `clean_tabular` (right-edge labels, left-closed bins). H4 and daily blocks end at 17:00 New York. Event tests use the last completed bar whose right edge is at or before the release time, so no bar contains post-release minutes.
- **Signals.** `causal_signals` at zone 5 (a 40-bar window), with the 3.14e-9 sentinel rows dropped.
- **Size test.** Walk-forward: the model for each year from 2016 to 2026 is trained only on earlier years, and compared with trailing volatility and hour of day alone.
- **Sealed data.** August to October 2026 stays untouched for one final look with a single pre-registered rule. All signals stop at July 2026.
- **Two bugs caught and fixed.** A look-ahead leak (a bar containing post-release minutes made M16 look like 63%; it is 51% after the fix) and a bar-offset merge error for horizons that do not divide the day, which had dropped one year of volatility data.
- **Data.** H1 history for 2012-01 to 2026-07 (175 monthly files) pulled with `research/pull_history.py --granularity H1`. Rates files from FRED (DGS10, DGS2) and the Japan Ministry of Finance JGB history are in `data/rates`. The puller now takes `--instrument`, `--granularity`, `--start` and `--end`, with tests for S5 and H1 chunking (40 pass).

What was built

Item	What it does
`research/pull_history.py`	Pulls monthly OANDA candle files at S5, M1 or H1 with `--instrument`, `--granularity`, `--start`, `--end`. 40 tests pass
`research/signal_lab.py`	Causal B, S, Theta and Mu at any bar, next-bar scoring, walk-forward helpers
`research/regions.py`, `trials.csv`	Theta×Mu region lookup and learner; log of the zone 5–15 scan trials on M30 and M60
`events.csv`, `events_2023_2024.csv`	Release times for jobs, CPI, FOMC and BoJ: 62 events (2025–26) and 80 (2023–24)
`EVENT_RULES.md`, `LONG_HORIZON_RULES.md`	Rules frozen before the 2023–24 and H1/H4/daily tests, plus the recorded negative event result
`patches/`	`pull_history_s5`, `swing_scoring_pull_args` and `pull_history_h1`, applied to your repo

Decisions on record: trading infrequently (daily or swing) is acceptable, B, S, Theta and Mu stay together in the model, and USD/JPY comes first before other pairs.

The plan

Five steps, in order. Each step has to pass before the next one is worth doing.

1. **Pull history.** Fetch several months of USD_JPY bars from OANDA at the chosen intake, then the other pairs (MXN, CHF, EUR). Answers: is there enough independent data to test anything?
2. **Run the edge test.** Score each feature against the next N bars' return and volatility, net of the measured spread (median 1.6 pips, 0.016 yen), against a random-walk baseline and a simple baseline such as plain price momentum. Answers: do X and Y predict anything that plain price does not?
3. **Backtest the survivors.** Costed, walk-forward backtests with bid/ask fills. This is also where pyramiding and stop-loss rules get designed. Answers: does it make money after costs on data it was not fitted to?

4. **Forward-test on demo.** Small size on the demo account. Answers: do live slippage, missed fills and data delays break it?
5. **Live.** Only after the trading rewrite, authentication on the routes, and an approved OANDA live account.

Hypotheses to test first

Write the expectation down before running each test, so a result cannot be reinterpreted afterwards.

#	Hypothesis	Why it might be true	What would kill it
H1	The squared X−Y gap (old Margin) predicts the next bars' volatility	A squared quantity relates to volatility, and the gap measures disagreement between X and Y	No relation to forward volatility beyond what the random-walk baseline gives
H2	sign(X − Y) combined with the gap predicts direction	The gap alone has no direction, so a sign must come from somewhere	Hit rate within noise of 50% after the spread
H3	Current B, S, Theta or Mu predict forward return	They are the live signal, so they have to be tested too	Correlation with forward return indistinguishable from the random-walk baseline
H4	X and Y add information beyond plain price momentum	The reason to use two inputs instead of one	A simple momentum or mean-reversion baseline does as well
H5	Engineered X, Y, B and S make B, S, Theta and Mu each predict the next horizon's direction above 55% out of sample	55% is the stated goal. It is the acceptance test, not a design target, and it has to hold on August to October data that no design choice saw	Out-of-sample hit rate under 55%, or above 55% but under the break-even hit rate for the spread at that horizon (about 59% at M60 and higher at shorter horizons)

Testing rules

A feature only counts if it beats these baselines on data it was not fitted to.

- **Random-walk baseline.** Run every feature on random walks with random volume (the test above is the template), and keep only what real data beats clearly.
- **Simple baseline.** Compare against plain price momentum and mean reversion. If X and Y do not beat them, they add nothing.
- **Costs always in.** Judge every result net of the spread (median 1.6 pips, about $0.10 per 1005-unit round trip on USD_JPY, but about 4.8 pips around the 21:00 to 22:00 UTC rollover), with fills at bid and ask.
- **Walk-forward only.** Fit on one stretch, test on a later one, and never tune on the test stretch.
- **Overlap.** The saved windows overlap heavily, so a month of them holds far fewer independent samples than it looks. Count independent bars, not rows.
- **Count every trial.** Log each variant tried. Enough variants will always produce one that looks good by luck, so the number of trials decides how much a good result is worth.

Where this leaves the project

Price, volume, the calendar and the rate spread do not predict USD/JPY direction in this data, so the next step is a choice of target, not another indicator. Standing rules are unchanged: `ALLOW_LIVE` stays False, August to October 2026 stays sealed, and B, S, Theta and Mu stay in the model.

Option	What it tests	Cost and odds
Event gate on H1 and H4	Mu contrarian taken only on the first bar after a jobs, CPI, FOMC or BoJ release	One run. About 140 releases (2023–2026), so at most a lead. Expected flat, since the M1–M30 version failed on 2023–24
Markets with real volume (BTC/USD, then SPY)	The same frozen B−S>0 and Mu contrarian rules. X is signed tick volume, and forex tick counts are a weak stand-in for traded volume	Needs an exchange data download and a small loader. A reproducible 52–53% would be a real result; 55% would be unusual

Option	What it tests	Cost and odds
Ranking many assets	Long the top-scored, short the bottom-scored instruments, which removes the market-wide drift	Needs 20–50 instruments and a cross-sectional test
Volatility forecasting (SPY project)	Whether the Theta and S size signal helps a volatility model	Trailing volatility does nearly all the work, so expect a small gain

Decided against for now: adding more price or volume indicators (same information, more chances of a fluke), a carry test (outside the goal of predicting direction), and a live forward-logger (low odds of finding what 15 years of history missed, though it would give a clean record). The yield level and rate-surprise versions of the rate idea are untested; only the 5-day change in the spread was tried.

Open questions and parked items

- [ ] Which intake and horizon to pull history at? The 2025 files used 60 s and 114 s bars.
- [ ] How far back to pull, and for which pairs first? USD_JPY is the only one wired end to end.
- [ ] Add a `LEGACY_S` switch to `transform_signal.py` so the 2025 S can be studied beside the current one? Off by default, and checked by reproducing the saved 2025 files exactly.
- [ ] OANDA live account approval (item 14): per-mode keys, fail-closed Live and the `divmod` trader split wait on this.
- [ ] Parked until the signal shows an edge: pyramiding up to `ZONE*BLOTSIZE`, stop loss on the OANDA side, drawdown limit, USD exposure cap, a real `cancel_order`, duplicate-order protection, and authentication on the routes.
- [ ] Run the event gate on H1 and H4 to close the calendar idea, then write up the result here.
- [ ] Which target next: real-volume markets (BTC/USD, SPY), cross-sectional ranking, or the SPY volatility project?
- [ ] If BTC/USD: which exchange feed, and does the bar boundary stay at midnight UTC for a 24/7 market?

- [ ] Test the yield level and rate surprises, not only the 5-day change in the US–Japan spread?

Experiment log

One row per test run, newest first. Add a row before running a test, then fill in the result.

Date	Test	Result	Decision
2026-11-10	2025 sawtooth on 30 random walks vs real Sept 25 bars (305 bars)	Same sawtooth on noise: skew -1.44 vs -1.28 on real data, sharp drops 14.3% vs 13.2%	The shape is an artifact of squaring and z-scoring, so it is not evidence of signal. Test the gap as a volatility feature (H1) instead
2026-11 (earlier)	Price-only direction on M1–M30 bars, 2023–2026: B–S>0, Mu contrarian, Theta×Mu regions, time-of-day and volatility cells, logistic regression, clusters	About 50% out of sample. The 2026 "52–53%" matched the 52.4% share of up bars that year, plus best-of-many selection	No direction edge from price alone at M1–M30
2026-11 (earlier)	Do Theta, S, Mu predict move size beyond trailing volatility and hour of day? (M1–M30)	Theta and S add 0.03–0.19 points of R^2; Mu adds none	Small volatility-size signal, not tradable alone
2026-11 (earlier)	Event windows (jobs, CPI, FOMC, BoJ; 62 events 2025–26, 80 events 2023–24)	Moves 2–4× normal range and event type predicts size. Post-release drift flipped sign: fade worked 2025–26, continuation 2023–24	Event timing predicts size, not direction

Date	Test	Result	Decision
2026-11 (earlier)	Calendar-gated Mu contrarian, M1–M30, 2025–26	55–61% at about six horizons (M1/25m, M25/25m, M26/25m, M28/25m, M30/25m, M30/55m). The first run of all 30 horizons leaked look-ahead (M16 showed 63%, 51% after the fix)	Shortlist frozen in EVENT_RULES.md and treated as a lead only
2026-11 (earlier)	Pre-registered test of that shortlist on 2023–24	No rule passed	Lead rejected
2026-11-12	B−S>0 and Mu contrarian on H1, H4, daily, 2012 to July 2026 (90,092, 22,642 and 3,818 signals)	Hit rates 48.7–51.2% against always-up 51.4% (H1), 52.4% (H4), 52.9% (D). Every rule at or below always-up. Net of the 1.6-pip spread: −0.6 to −1.7 pips per trade	No lead at any horizon
2026-11-12	Move-size test on H1 and H4, walk-forward 2016–2026	Trailing volatility and hour explain 12.4% (H1) and 7.7% (H4) of next-bar size. B, S, Theta, Mu add 0.07 and 0.15 points, positive in 8 of 11 and 10 of 11 years. Theta rank correlation about +0.02	Real but too small to use
2026-11-12	Yield spread (US 10y minus Japan 10y, 5-day change, lagged one day), alone and as a gate on Mu contrarian, H1/H4/D	Spread alone hit 50.0%, 50.0%, 48.8% against always-up 51.4%, 52.4%, 52.9%. Gating Mu left 50.8% (H1, H4) and 50.0% (D)	No predictive power for direction